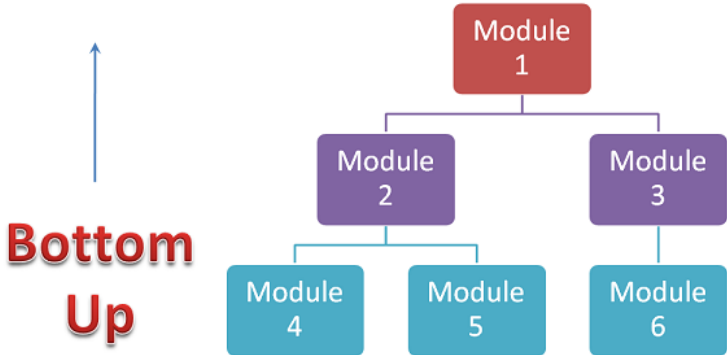


FC5463 Software Testing Winter 2022 Solution

Government Polytechnic, Amravati (An Autonomous Institute of Government of Maharashtra) Brief Solution and Structured marking Scheme Course Code and Name: FC5463 Software Testing	
Q.1. Attempt any TWO.	Max Marks & (marking scheme)
a) Explain equivalence partitioning with suitable example.	4
- Equivalence Class Partitioning is type of black box testing technique which can be applied to all levels of software testing like unit, integration, system, etc. - In this technique, input data units are divided into equivalent partitions that can be used to derive test cases which reduces time required for testing because of small number of test cases. - It divides the input data of software into different equivalence data classes. - You can apply this technique, where there is a range in the input field. Example: - Let's consider the behavior of Order Pizza Text Box Below - Pizza values 1 to 10 is considered valid. A success message is shown. - While value 11 to 99 are considered invalid for order and an error message will appear, "Only 10 Pizza can be ordered" - Here is the test condition - Any Number greater than 10 entered in the Order Pizza field(let say 11) is considered invalid. - Any Number less than 1 that is 0 or below, then it is considered invalid. - Numbers 1 to 10 are considered valid - Any 3 Digit Number say -100 is invalid.	2 marks each for explain and example
b) Describe the terms: i) White Box Testing	4
i) White Box Testing - software testing technique in which internal structure, design and coding of software are tested to verify flow of input-output and to improve design, usability and security. - In white box testing, code is visible to testers so it is also called Clear box testing, Open box testing, Transparent box testing, Code-based testing and Glass box testing. - One of the basic goals of white box testing is to verify a working flow for an application. ii) Black Box Testing - A method in which the functionalities of software applications are tested without having knowledge of internal code structure, implementation details and internal paths. - Mainly focuses on input and output of software applications - It is entirely based on software requirements and specifications. 	2 marks each

c) List any four objectives of software testing.	4
• Verification:- allows testers to confirm that the software meets the various business and technical requirements stated by the client before the inception of the whole project. • Validation:- Confirms that the software performs as expected and as per the requirements of the clients. • Defects:- to find different defects in the software to prevent its failure or crash during implementation or go live of the project. • Providing Information:- during the process of software testing, testers can accumulate a variety of information related to the software and the steps taken to prevent its failure • Quality Analysis:- Testing helps improve the quality of the software by constantly measuring and verifying its design and coding. • Compatibility:- It helps validate application's compatibility with the implementation environment, various devices, Operating Systems, user requirements, among other things. • Verifying Performance & Functionality:- It ensures that the software has superior performance and functionality.	Any 4 (1 mark each)
Q.2.(A) Attempt any THREE.	
a) Recall the term integration testing.	2
• Aims to test different parts of the system in combination in order to assess if they work correctly together • By testing the units in groups, any faults in the way they interact together can be identified. • Testers can adopt either a bottom-up or a top-down integration method.	1 mark each
b) Summarize bottom up integration testing.	2
• A strategy in which the lower level modules are tested first. • These tested modules are then further used to facilitate the testing of higher level modules. • The process continues until all modules at top level are tested.  @guru99.com	1 mark for concept and 1 for diagram

c) Recall the term: i) stub ii) drivers	2
STUBS: - Assume you have 3 modules, Module A, Module B and module C. Module A is ready and we need to test it, but module A calls functions from Module B and C which are not ready, so developer will write a dummy module which simulates B and C and returns values to module A. This dummy module code is known as stub. DRIVERS: - Now suppose you have modules B and C ready but module A which calls functions from module B and C is not ready so developer will write a dummy piece of code for module A which will return values to module B and C. This dummy piece of code is known as driver.	1 mark each
d) Recall the term regression testing.	2
- REGRESSION TESTING is defined as a type of software testing to confirm that a recent program or code change has not adversely affected existing features. - Regression Testing is nothing but a full or partial selection of already executed test cases which are re-executed to ensure existing functionalities work fine. - This testing is done to make sure that new code changes should not have side effects on the existing functionalities. It ensures that the old code still works once the latest code changes are done.	
Q.2.(B) Attempt any TWO.	
a) Describe the term performance testing.	4
- The goal of performance testing is to evaluate the application's performance with respect to real world scenarios. The following issues must be addressed during performance testing: (i) Performance of the system during peak hours (response time, reliability and availability). (ii) Points at which the system performance degrades or system fails. (iii) Impact of the degraded performance on the customer loyalty, sales and profits. There are two types of performance testing Load Testing and stress testing	1 mark each point
b) Summarize the term security testing.	4
- Security is the procedure used to protect information from various threats. - The user wants implementation of a safeguard to protect personal, sensitive and financial information. - We want data to be accurate, reliable and protected against unauthorized access. - Security involves various threats such as unauthorized users, malicious users, message sent to an unintended user, etc.	1 mark each point

c) Compare Alpha testing with Beta testing.		4
Alpha Testing	Beta Testing	Any 4 points 1 mark each
Alpha Testing is done by testers who reside internally as employees of that company or organization.	Users and clients do it, and they are not the employees of any company or organization.	
This testing type is carried out at the developer's site.	This testing is carried at the end user's location.	
In-depth reliability and security of the product are not done in this testing type.	In-depth reliability and security of the product are checked, and patches are being released.	
It uses both white box as well as black box testing.	It uses black box Testing only.	
Execution of the test cycle might take a longer time.	Execution time for the test requires only a few months (at maximum).	
Q.3.(A) Attempt any THREE.		
a) Describe how to set up criteria for testing.		2
- Encountering more than a certain number of defects, causing frequent stoppage of testing activity; - Hitting show stoppers that prevent further progress of testing (for example, if a database does not start, further tests of query , data manipulation, and so on are simply not possible to execute); and - Developers releasing a new version which they advice should be used in instead of the product under test (because of some critical defect fixes.) When such conditions are addressed, the tests can resume.		
b) Enlist the contents in test plan.		2
Test strategy, Test objectives, Test schedule, Test estimation, deliverables, and resources required, what needs to be tested, how testing is going to be done ,time line, risks involved		Any 4 ½ mark each
c) Enlist test deliverable during testing & after testing.		2
Test deliverables are provided during the testing - Test Scripts - Simulators. - Test Data - Test Traceability Matrix - Error logs and execution logs. Test deliverables are provided after the testing cycles is over. - Test Results/reports - Defect Report - Installation/ Test procedures guidelines - Release notes		1 mark each

d) Describe different human resources required for software testing and their tasks.			2			
No.	Member	Tasks	Any 2 1 mark each			
1.	Test Manager	Manage the whole project, Define project directions Acquire appropriate resources				
2.	Tester	Identifying and describing appropriate test techniques/tools/ automation architecture				
3.	Developer in Test	Implement the test cases, test program, test suite etc.				
4.	Test Administrator	Builds up and ensures Test Environment and assets are managed and maintained Support Tester to use the test environment for test execution				
5.	SQA members	Take in charge of quality assurance Check to confirm whether the testing process is meeting specified requirements				
Q3 (B) Attempt any ONE						
a) Design any 4 test cases for testing following Registration form.						
ID code Name Email Mobile Submit Reset						
Sr. No.	Test Case ID	Testcase Objectives	Prerequisite	Steps	Input Data	Expected Result
1	TC01	Verify that the form is visible properly	Application should be accessible	Opening the application.	All fields data to be entered	Able to enter all the data

2	TC02	Verify that the email entered is valid.	Application should be accessible	Enter Valid/Invalid emails	Valid and invalid email address	Valid mail is accepted	
3	TC03	Verify that submit button works properly	Application should be accessible	Enter all data and click on submit button	Valid data	Form is submitted successfully	
4	TC04	Verify that reset button works properly	Application must be opened.	Enter all data/ certain fields and click on reset	Valid data	Form should reset after click on reset button	
Q 3 b) Identify activities involved in test people management.							
1) People management requires the ability to hire, motivate and retain the right people. These skills are seldom formally taught. (unlike technical skills.) 2) Project manager often learn these skills in a “sink or swim” mode, being thrown head-on into the task. 3) These team-building exercises should be conducted in ongoing and sustained mode, rather than be done in one burst. 4) The effort of these exercises tends to wear out under the pressure of deadlines of delivery and quality. 5) The common goals and the spirit of teamwork have to be internalized by all stakeholders. Such an internalization and upfront team building has to be part of the planning process the team to succeed.							Any 4 points 1 mark each
Q.4.(A) Attempt any TWO.							
a) Recall the term defect. - A Defect in Software Testing is a variation or deviation of the software application from end user’s requirements or original business requirements. - A software defect is an error in coding which causes incorrect or unexpected results from a software program which does not meet actual requirements.							2 marks
b) State the purpose of preparing defect report. The purpose of a defect report is to state the problem as clearly as possible so that developers can replicate the defect easily and fix it.							

c) Recall the term priority of defect. • Priority is defined as the order in which the defects should be resolved. • The priority status is usually set by the QA team while raising the defect against the dev team mentioning the timeframe to fix the defect. • The Priority status is set based on the requirements of the end users.	
Q 4 (B) Attempt any TWO.	
a. Describe defect management process with neat labelled diagram. • Defect Management is a systematic process to identify and fix bugs.  <pre> graph TD Discovery[Discovery] --> Categorization[Categorization] Categorization --> Resolution[Resolution] Resolution --> Verification[Verification] Verification --> Closure[Closure] Closure --> Reporting[Reporting] Reporting --> Discovery </pre> • In the discovery phase, the project teams have to discover as many defects as possible, before the end customer can discover it. • A defect is said to be discovered and change to status accepted when it is acknowledged and accepted by the developers. • Defect categorization help the software developers to prioritize their tasks • Defect Resolution in software testing is a step by step process of fixing the defects. • Once a defect has been resolved and verified, the defect is changed status as closed. • Defect Reporting in software testing is a process in which test managers prepare and send the defect report to the management team for feedback on defect management process and defects' status.	
b) Identify severity & priorities of following bugs. i) Incorrect logo on website Severity – Low Priority- High ii) Defect in generating train PNR number Severity – Critical Priority- High iii) Not able to view tickets booked in last year Severity – Medium Priority- Medium	1 Mark each

iv) Background colour of the 'submit' button is not matches with 'reset' button Severity – Low Priority- Low																																									
c) Design sample bug report based on following data: Project Name: IRCTC Product Name: IRCTC website Release version : 1:0 List of bugs: i) Not able to book tickets for next month <table border="1" data-bbox="282 784 1233 2002"> <tr> <td>ID</td><td>DI</td></tr> <tr> <td>Project</td><td>IRCTC</td></tr> <tr> <td>Product</td><td>IRCTC website</td></tr> <tr> <td>Release Version</td><td>1.0</td></tr> <tr> <td>Module</td><td>Ticket_book</td></tr> <tr> <td>Detected Build Version</td><td>1.1</td></tr> <tr> <td>Summary</td><td>Not able to book tickets for next month</td></tr> <tr> <td>Description</td><td>When we select booking date, max day we can select is last day of current month. Next month onwards we cannot book tickets.</td></tr> <tr> <td>Steps to Replicate</td><td>Select Boarding Station (From) and Destination Station (To) Select Date Click on search (disabled as date is from next month)</td></tr> <tr> <td>Actual Result</td><td>Search button is disabled.</td></tr> <tr> <td>Expected Results</td><td>After clicking Search button, trains between stations should have been shown.</td></tr> <tr> <td>Attachments</td><td>Screenshots</td></tr> <tr> <td>Remarks</td><td>NA</td></tr> <tr> <td>Defect Probability</td><td>80%</td></tr> <tr> <td>Defect Severity</td><td>Critical</td></tr> <tr> <td>Defect Priority</td><td>High</td></tr> <tr> <td>Reported By</td><td>XYZ</td></tr> <tr> <td>Assigned To</td><td>ABC</td></tr> <tr> <td>Status</td><td>Opened</td></tr> <tr> <td>Fixed Build Version</td><td>1.0</td></tr> </table>	ID	DI	Project	IRCTC	Product	IRCTC website	Release Version	1.0	Module	Ticket_book	Detected Build Version	1.1	Summary	Not able to book tickets for next month	Description	When we select booking date, max day we can select is last day of current month. Next month onwards we cannot book tickets.	Steps to Replicate	Select Boarding Station (From) and Destination Station (To) Select Date Click on search (disabled as date is from next month)	Actual Result	Search button is disabled.	Expected Results	After clicking Search button, trains between stations should have been shown.	Attachments	Screenshots	Remarks	NA	Defect Probability	80%	Defect Severity	Critical	Defect Priority	High	Reported By	XYZ	Assigned To	ABC	Status	Opened	Fixed Build Version	1.0	1 Mark each
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ii) Logo not displayed properly

ID	D2
Project	IRCTC
Product	IRCTC website
Release Version	1.0
Module	Home Page
Detected Build Version	1.0
Summary	Logo not displayed properly
Description	When we go to the site, logo is not displayed.
Steps to Replicate	Type the URL and press Enter
Actual Result	On Home Page Logo is not properly placed
Expected Results	Logo on Home page should be properly displayed
Attachments	Screenshots
Remarks	NA
Defect Probability	100%
Defect Severity	Low
Defect Priority	High
Reported By	XYZ
Assigned To	ABC
Status	Opened
Fixed Build Version	1.0

iii) Not able to add more than 3 passengers

ID	D3
Project	IRCTC
Product	IRCTC website
Release Version	1.1
Module	AddPassengers
Detected Build Version	1.1.0
Summary	Not able to add more than 3 passengers

Description	While adding passenger details only 3 passenger details can be added.
Steps to Replicate	Search trains between Source and destination click on book now add passenger details
Actual Result	Upto 3 passengers can be added
Expected Results	At least 4 passengers should have been added
Attachments	Screenshots
Remarks	NA
Defect Probability	100%
Defect Severity	Major
Defect Priority	Medium
Reported By	XYZ
Assigned To	ABC
Status	Opened
Fixed Build Version	1.0

iv) PNR (Ticket Number) number generated is duplicate.

ID	D3
Project	IRCTC
Product	IRCTC website
Release Version	1.1
Module	BookTicket
Detected Build Version	1.1.0
Summary	PNR (Ticket Number) number generated is duplicate.
Description	Same PNR number is getting generated for two different logins.
Steps to Replicate	Search trains between Source and destination click on book now book ticket make payment
Actual Result	Ticket gets booked and duplicate PNR is generated.
Expected Results	PNR for two different tickets should be different.
Attachments	Screenshots
Remarks	NA

	<table><tr><td>Defect Probability</td><td>100%</td></tr><tr><td>Defect Severity</td><td>Critical</td></tr><tr><td>Defect Priority</td><td>High</td></tr><tr><td>Reported By</td><td>XYZ</td></tr><tr><td>Assigned To</td><td>ABC</td></tr><tr><td>Status</td><td>Opened</td></tr><tr><td>Fixed Build Version</td><td>1.0</td></tr></table>	Defect Probability	100%	Defect Severity	Critical	Defect Priority	High	Reported By	XYZ	Assigned To	ABC	Status	Opened	Fixed Build Version	1.0	
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Q.5(A) Attempt any TWO.																
a. Recall the term Automation Testing. Automation Testing is a software testing technique that performs using special automated testing software tools to execute a test case suite.		2 marks for definition														
b. Compare automation testing with manual testing (any 2 points)		2 marks for any two points														
Automation Testing	Manual Testing															
Perform the same operation each time	Test execution is not accurate all the time. Hence not reliable.															
Useful to execute the set of test cases frequently.	Useful when the test case only needs to run once or twice.															
Fewer testers required to execute the test cases.	Large number of tester required															
Testers can fetch complicated information from code.	Does not involve in programming task to fetch hidden information.															
Faster	Slow															
Not Helpful in UI	Helpful in UI															
High Cost	Less than automation.															
c. Solve the following to calculate test design coverage. Total No of Test Cases: 240 No of requirements : 70 Number of requirements mapped to test cases: 60 Ans: Test Design Coverage = ((Total number of requirements mapped to test cases) / (Total number of requirements))*100 Test Design Coverage= 60/70*100 = 85.714		1 mark for formula and 1 for calucaltion														

5 (B) Attempt any TWO.	
a. Describe any four process metrics with suitable example. 1. Test Case Preparation Productivity - It is used to calculate the number of Test Cases prepared and the effort spent for the preparation of Test Cases. Formula: - Test Case Preparation Productivity = (No of Test Case)/ (Effort spent for Test Case Preparation) - E.g.: - No. of Test cases = 240 - Effort spent for Test case preparation (in hours) = 10 - Test Case preparation productivity = 240/10 = 24 test cases/hour 2. %ge Test cases Executed: This metric is used to obtain the execution status of the test cases in terms of %ge. %ge Test cases Executed = (No. of Test cases executed / Total no. of Test cases written) * 100. 3. %ge Test cases not executed: This metric is used to obtain the pending execution status of the test cases in terms of %ge. %ge Test cases not executed = (No. of Test cases not executed / Total no. of Test cases written) * 100. 4. %ge Test cases Passed: This metric is used to obtain the Pass %ge of the executed test cases. %ge Test cases Passed = (No. of Test cases Passed / Total no. of Test cases Executed) * 100.	1 for each
b. Summarize factors considered for a selecting a testing tool. 1. Meeting requirements- There are plenty of tools available in the market but rarely do they meet all the requirements of a given product or a given organization. Evaluating different tools for different requirements involve significant effort, money, and time. 2. Technology expectations Test tools in general may not allow test developers to extends/modify the functionality of the framework. So extending the functionality requires going back to the tool vendor and involves additional cost and effort. A good number of test tools require their libraries to be linked with product binaries. 3. Training/skills- While test tools require plenty of training, very few vendors provide the training to the required level. Organization level training is needed to deploy the test tools, as the user of the	1 for each

test suite are not only the test team but also the development team and other areas like configuration management.	
4. Management aspects- A test tool increases the system requirement and requires the hardware and software to be upgraded. This increases the cost of the already- expensive test tool.	
c. Identify the need of automation testing by giving suitable example. • Manual Testing of all workflows, all fields, all negative scenarios is time and money consuming • It is difficult to test for multilingual sites manually • Test Automation in software testing does not require Human intervention. You can run automated test unattended (overnight) • Test Automation increases the speed of test execution • Automation helps increase Test Coverage • Manual Testing can become boring and hence error-prone.	3 marks for explanation and 1 for example
6 (A) Attempt any THREE. a. outline why webdriver is better than selenium RC. • The WebDriver proves itself to be better than both Selenium IDE and Selenium RC in many aspects. • It implements a more modern and stable approach in automating the browser's actions. • WebDriver, unlike Selenium RC, does not rely on JavaScript for Selenium Automation Testing. • It controls the browser by directly communicating with it.	2 Marks
b. Enlist the languages supported by selenium webdriver. • Java • C# • PHP • Python • Perl • Ruby	0.5 for one language
c. Recall any four features of metrics bug tracker. • Email notifications: It sends out emails of updates, comments, resolutions to the concerned stakeholders. • Access Control: You can control user access at a project level • Customize: You can easily customize Mantis as per your requirements. • Mobile Support: Mantis supports iPhone, Android, and Windows Phone Platforms. • Plugins: An ever-expanding library of plugins to add custom functionality to Mantis Issue Tracker.	0.5 for each

d. Describe the term performance automation by giving suitable example.	
B. Attempt any ONE.	
a. Design a selenium script to automate login form. <pre> import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.testng.Assert; import org.testng.annotations.Test; public class LoginAutomation { public void login() { System.setProperty("webdriver.chrome.driver", "path of driver"); WebDriver driver=new ChromeDriver(); driver.manage().window().maximize(); driver.get("https://www.browserstack.com/users/sign_in"); WebElement username=driver.findElement(By.id("user_email_Login")); WebElement password=driver.findElement(By.id("user_password")); WebElement login=driver.findElement(By.name("commit")); username.sendKeys("abc@gmail.com"); password.sendKeys("your_password"); login.click(); } } </pre>	6 marks
b. Design a selenium script to automate following form. <pre> import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.testng.Assert; public class FormAutomation { public void login() { System.setProperty("webdriver.chrome.driver", "path of driver"); WebDriver driver=new ChromeDriver(); driver.manage().window().maximize(); driver.get("https://www.browserstack.com/users/sign_in"); WebElement name=driver.findElement(By.id("name")); WebElement ref_ID=driver.findElement(By.id("referral_id")); WebElement continue=driver.findElement(By.name("continue")); name.sendKeys("abc"); ref_ID.sendKeys("ABC1212"); continue.click();} } </pre>	6 marks